CCNA / CCNP QUICK REFERENCE

Cisco IOS Commands Cheat Sheet

Comprehensive CLI Guide with Domain Name, SSH v2, User Auth, Auto-MDIX, Interface Diagnostics & VLAN Deletion / Factory Reset

EXEC MODES & NAVIGATION

User EXEC Mode	Router>
Privileged EXEC Mode	enable
Global Configuration	configure terminal
Exit Mode / End Config	exit / end
Context Help	? or command ?

DOMAIN, SSH V2 & USER AUTHENTICATION

Security Best Practice: Always enforce SSH Version 2 (ip ssh version 2) as SSHv1 contains known security vulnerabilities. Generating an RSA key modulus of at least 2048 bits is required for SSHv2.

1. Set Hostname	hostname <name>
2. Set IP Domain Name	ip domain-name <domain.com>
3. Generate RSA Keys	crypto key generate rsa general-keys modulus 2048
4. Enforce SSH Version 2	ip ssh version 2
5. Create Local User	username <user> secret <pass>
6. Enable Local Auth on VTY	line vty 0 4 login local
7. Restrict VTY to SSH	transport input ssh
Verify SSH Status	show ip ssh | show ssh

GLOBAL CONFIG & SECURITY

Encrypted Enable Secret	enable secret <pass>
Console Port Password	line con 0 password <pass> | login
Encrypt Passwords	service password-encryption
Disable IP Domain Lookup	no ip domain-lookup
Save Configuration	copy run start

INTERFACE SPEED, DUPLEX & AUTO-MDIX

Select Interface / Range	interface fa0/1 interface range fa0/1 - 12
Configure Port Speed	speed auto | 10 | 100 | 1000
Configure Duplex Mode	duplex auto | full | half
Enable Auto-MDIX	mdix auto
Enable / Shutdown Port	no shutdown | shutdown

Auto-MDIX Rule: Automatic MDIX cable detection requires both speed auto and duplex auto to be active on the interface.

VLANS & TRUNKING

Create / Name VLAN	`vlan <id>` `name <vlan_name>`
Assign Access VLAN	`switchport mode access` `switchport access vlan <id>`
Assign Voice VLAN	`switchport voice vlan <id>`
Trust QoS CoS Markings (Voice)	`mls qos trust cos`
Remove VLAN Assignment from Port	`no switchport access vlan`
Configure Trunk Port	`switchport mode trunk`
Allowed Trunk VLANs	`switchport trunk allowed vlan <list>`

Tip — 'do' Command: Use the do prefix to run privileged EXEC commands (e.g. `do show vlan brief`) without leaving configuration mode.

ROUTER-ON-A-STICK (TRUNK SUBINTERFACES)

Select Subinterface	`interface <phys_intf>.<subintf_id>` `e.g. interface G0/0/1.10`
Add Description	`description <text>`
Tag VLAN (802.1Q Encapsulation)	`encapsulation dot1Q <vlan_id>`
Assign Gateway IP Address	`ip address <ip> <subnet_mask>`
Return to Global Config	`exit`

Enable Physical Trunk Interface	`interface <phys_intf>`
	`description Trunk link to <switch>`
	`no shutdown`

Router-on-a-Stick: Create one subinterface per VLAN on the router's physical interface connected to the switch trunk port. Each subinterface is tagged with `encapsulation dot1Q <vlan_id>` and given an IP address that serves as the default gateway for that VLAN. The parent physical interface carries no IP address of its own and must be enabled with `no shutdown`.

Example — 3 VLANs on G0/0/1:

```
interface G0/0/1.10
  description Default Gateway for VLAN 10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
  exit
interface G0/0/1.20
  description Default Gateway for VLAN 20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
  exit
interface G0/0/1.99
  description Default Gateway for VLAN 99 (Native/Mgmt)
  encapsulation dot1Q 99
  ip address 192.168.99.1 255.255.255.0
  exit
interface G0/0/1
  description Trunk link to S1
  no shutdown
```

MAC ADDRESS TABLE MANAGEMENT

Add Static MAC Address	`mac address-table static <mac_addr> vlan <id> interface <intf>`

Remove Static MAC Address	no mac address-table static <mac_addr> vlan <id>
Set Dynamic Aging Time	mac address-table aging-time <seconds>
Disable MAC Aging (per VLAN)	mac address-table aging-time 0 vlan <id>
Clear Dynamic Entries (All)	clear mac address-table dynamic
Clear Dynamic Entries (Interface)	clear mac address-table dynamic interface <intf>
Clear Dynamic Entries (VLAN)	clear mac address-table dynamic vlan <id>

Static MAC Addresses: A static entry permanently binds a specific host's MAC address to a switch port and VLAN, so that host is only ever learned on that port — useful for servers, printers, or security-sensitive devices. Static entries do not age out and survive normal MAC table refreshes, but are lost on reload unless the configuration is saved with copy run start.

Example:

```
S1(config)# mac address-table static 000A.000B.000C vlan 10 interface fa0/6
S1(config)# do show mac address-table static
```

Show Full MAC Address Table	show mac address-table
Show Static Entries Only	show mac address-table static
Show Dynamic Entries Only	show mac address-table dynamic
Show Entries for a VLAN	show mac address-table vlan <id>
Show Entries for an Interface	show mac address-table interface <intf>
Show Current Aging Time	show mac address-table aging-time

VLAN DELETION & FACTORY RESET

Remove a Single VLAN	no vlan <vlan-id>
Delete Entire vlan.dat File	delete flash:vlan.dat
Delete vlan.dat (abbreviated, default location only)	delete vlan.dat
Erase Startup Config	erase startup-config
Reload Switch	reload

CAUTION — Reassign Ports Before Deleting a VLAN

The no vlan <vlan-id> global configuration mode command removes a VLAN from the switch's vlan.dat file. Before deleting a VLAN, reassign all member ports to a different VLAN first. Any ports that are not moved to an active VLAN are unable to communicate with other hosts after the VLAN is deleted, and will remain unable to do so until they are assigned to an active VLAN.

VLAN.DAT DELETION — Factory Default VLAN Config

The entire vlan.dat file can be deleted using the delete flash:vlan.dat privileged EXEC mode command. The abbreviated command version (delete vlan.dat) can be used if the vlan.dat file has not been moved from its default location. After issuing this command and reloading the switch, any previously configured VLANs are no longer present — this effectively places the switch into its factory default condition with regard to VLAN configurations.

NOTE — Full Factory Reset Procedure

To restore a Catalyst switch to its factory default condition: unplug all cables except the console and power cable from the switch, then enter the erase startup-config privileged EXEC mode command, followed by the delete vlan.dat command. Reload the switch to complete the reset.

VERIFICATION & DIAGNOSTICS (SHOW)

show ip interface brief	IP & status summary for all interfaces.

show interfaces fa0/1	Link status, speed, duplex, & error counters.
show interfaces status	Switchport mode, speed, duplex, & VLAN.
show controllers ... phy | inc MDIX	Verifies if Auto-MDIX is On or Off.
show running-config	Displays active running configuration in RAM.
show vlan brief	Lists all active VLANs and their assigned ports.
show vlan name <vlan_name>	Shows VLAN details filtered by VLAN name.
show vlan id <vlan_id>	Shows VLAN details filtered by VLAN ID number.
show vlan summary	Displays a count of existing, VTP, & extended VLANs.
show interface <intf> switchport	Detailed switchport mode, access VLAN, & voice VLAN info.
show mac address-table	MAC-to-port binding table.
clear counters fa0/1	Resets error counters on port Fa0/1.

MEDIA ERROR TROUBLESHOOTING

Diagnostic Guide for Port Counters:

Late Collisions (> 0)	Duplex mismatch (one end Full, one end Half) or Ethernet cable exceeding 100m.
CRC Errors (> 0)	Physical interference, faulty copper cabling, bad patch panel, or loose RJ-45 pins.
Runts (> 0)	Collisions on half-duplex links or a malfunctioning host NIC sending undersized frames (< 64 bytes).
Interface UP / Protocol DOWN	Layer 2 issue — speed/duplex mismatch, encapsulation, or VLAN assignment error.

IP ROUTING (STATIC & OSPF)

Static Route	`ip route <net> <mask> <next_hop>`
Default Route	`ip route 0.0.0.0 0.0.0.0 <next_hop>`
Enable OSPF	`router ospf <process_id>`
OSPF Network Statement	`network <ip> <wildcard> area <id>`

CLI SHORTCUTS & NAVIGATION

Tab	`Auto-completes entered CLI command.`
Up / Down Arrows	`Recalls previous command history.`
Ctrl + Shift + 6	`Aborts ping, traceroute, or DNS lookup.`
Ctrl + A / Ctrl + E	`Moves cursor to line start / end.`